Juan de Dios Alvarez

Data Analyst -- SQL, Dashboards and Financial Data Quality

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Profile

Data analyst and data scientist with seven years of industry work (2019 to present), four of them with US-based clients in a distributed, English-speaking team. I turn raw financial and operational data into SQL models, dashboards and written findings that non-technical teams use to decide: invoice, payment and bank reconciliation, credit-risk indicators at a development bank, and reports that run without manual steps. Each figure is traceable to its source. I use AI tools to automate repetitive parts of the work, with checks that catch their mistakes before anyone relies on the output.

Selected Results

- **Financial data quality:** designed a three-way accounts-payable reconciliation (proof of concept) linking tax e-invoices, payment complements and bank statement lines, with an exception taxonomy that separates timing differences from true discrepancies so an accountant reviews only actionable items.
- **Indicators for decision-makers:** at FIRA, a development bank, automated the data behind credit-assignment and non-payment-risk analysis: 74 scheduled extraction scripts over 9+ public sources, 60+ SQL functions, automated executive Excel reports and 15+ Apache Superset dashboards.
- **Dashboards for business teams:** Power BI dashboard that told a US client's sales team which products to offer each school district.
- **AI in the workflow, measured:** evaluated document-extraction methods (OCR, local vision and language models) head to head and adopted changes only on measured results (doubtful lines reduced from 175 to 77 on a fixed sample).

Professional Experience

Data Scientist & ML Engineer | Independent Practice

March 2026 -- present

- Designed and run in production the data system of a small retail business in Mexico: PostgreSQL warehouse (40+ tables) integrating point-of-sale, supplier, e-invoice and bank data, used daily by store staff.
- Derived monthly income-tax and VAT declaration inputs (taxable base, VAT charged, withholdings) directly from e-invoice XML, with month-by-month review.
- Audited every supplier invoice line against what reached the warehouse, sorting each gap by root cause; reconciled product identity across supplier catalogs, invoices and the point-of-sale system, which share no common key.
- Produced scheduled Excel reports on monthly sales and inventory value, each with notes on method and data coverage so business users can read them on their own.
- Built LLM pipelines with mechanical guardrails: schema-constrained outputs, rejection of text not present in the source, and local models for sensitive documents.

Economic Research Specialist (Data Scientist) | FIRA

January 2025 -- January 2026

- Replaced one analyst's manual Excel process, which fed credit-assignment and non-payment-risk analysis, with a scheduled, unattended pipeline: 74 extraction scripts (Python, API connections, Playwright) across 9+ public sources (INEGI, Banxico, SNIIM, SIAP, USDA) loading PostgreSQL with provenance metadata.
- Wrote 60+ PostgreSQL functions and materialized views computing indicators: inflation contributions, trade balances, supply-and-demand balances and crop yields.
- Delivered automated executive Excel reports and 15+ Apache Superset dashboards to decision-makers; set governance conventions (raw data preserved, lineage, GitLab) and supervised an undergraduate team.

- Built a crop-production estimation model for 310 plots from satellite time series (pandas, scikit-learn): Random Forest and Gradient Boosting compared by 5-fold cross-validation.

Data Scientist (contract) | CITEIM

October 2024 -- January 2025

- Built the dataset for a Mexican Sign Language recognition system from video (OCR, speech-to-text, MediaPipe keypoints).

Data Scientist & Team Lead | Punto Singular

December 2023 -- August 2024

- Led monthly analytics cycles for an EdTech client, from problem and metric definition to results presented against the agreed metric; KPI and quality dashboards in R, Python, Tableau and Looker used directly by business stakeholders; Power BI and data-flow design for LMS analytics.

Data Scientist (contract) | BairesDev

September 2023 -- November 2023

- Built a retrieval-augmented generation proof of concept for querying a legal client's contracts for obligations and risk clauses.

Data Scientist | D3Clarity (formerly Viqtor Davis North America)

February 2019 -- March 2023

- First data scientist of the company; defined the analytics roadmap and delivered it to US-based clients in a distributed team.
- Built executive and operational dashboards in Power BI and Tableau for non-technical decision-makers; extracted data from a master-data system into Power BI reports.
- Built forecasting models to identify consumer segments for client sales and marketing strategy; data lakes, ETL and master data management (Semarchy xDM).
- Classified public school-board meeting minutes with NLP on Azure ML, from proof of concept to production, feeding a Power BI dashboard for the client's sales team.

Earlier Experience

- Lecturer, Tecnologico de Monterrey Business School (KPI dashboarding in Tableau for economics and management students); HAWC observatory data analysis; data-acquisition software at CERN.

Education

- **Master's in Computer Science** -- INAOE, Mexico. Degree awarded; SEP professional license (cedula) 5916605.
- **PhD Candidate in Physics** -- University of Michoacan. Computational research: large-scale simulations of cosmic-ray air showers; principal author of two ICRC 2017 papers; co-author on 14 publications.
- **Bachelor's in Physics and Mathematics** -- University of Michoacan.
- **Certifications:** AWS Certified Cloud Practitioner; Semarchy xDM. **Languages:** Spanish (native), English (professional), Italian (basic-intermediate), French (basic).

Technical Skills

- **SQL and data:** PostgreSQL (PL/pgSQL, window functions, materialized views), MySQL, Firebird; ETL, REST API extraction, data lakes, master data management
- **BI and reporting:** Apache Superset, Power BI, Tableau, Looker, Excel (pivot tables, Power Query, automated reports)
- **Analysis and statistics:** Python (pandas, NumPy, scikit-learn), R; descriptive statistics, exploratory analysis, Random Forest, Gradient Boosting, PCA, forecasting, k-fold cross-validation, stratified sampling
- **AI tools:** LLM pipelines (Claude, local models via Ollama), schema-constrained extraction, OCR, RAG
- **Cloud and business systems:** AWS, Azure Machine Learning; SAP (end user)